Getting Started with .NET MAUI and Visual Studio 2022

David Ortinau

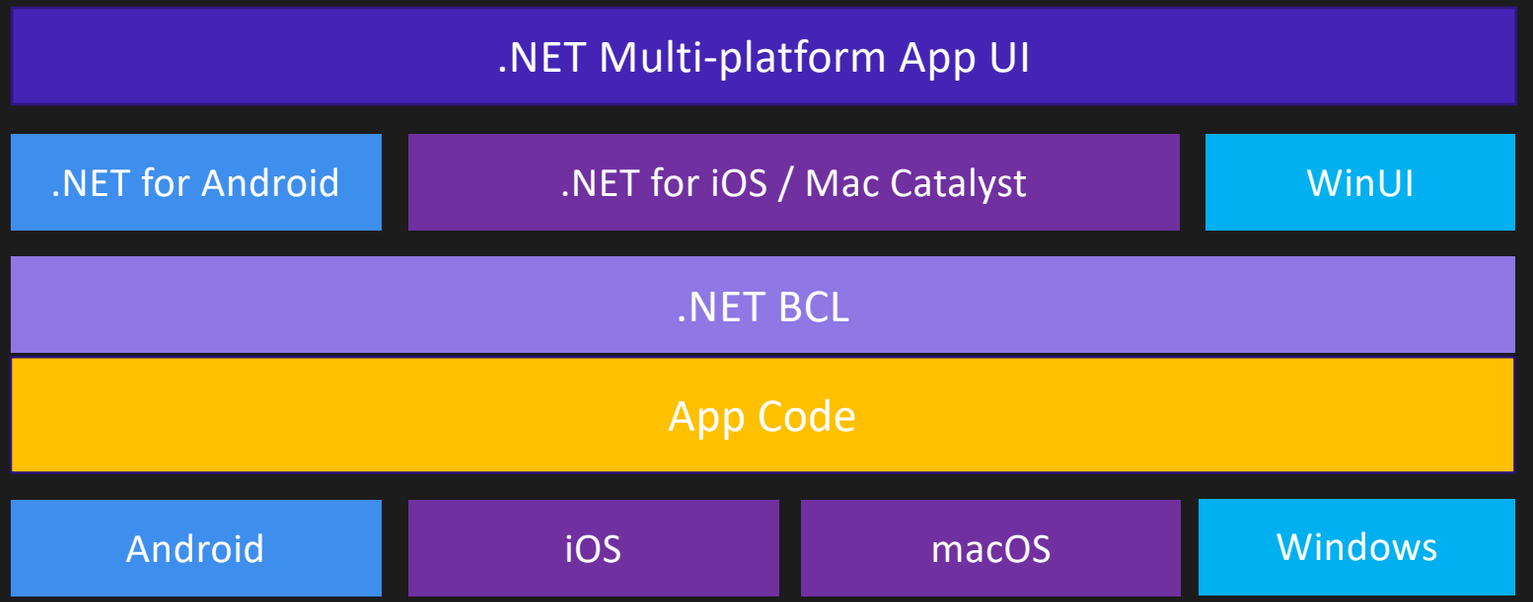
Principal Program Manager, .NET

david.ortinau@microsoft.com



.NET MAUI is the most productive way to develop native apps that perform great on any device that runs Android, iOS, macOS, or Windows from a single codebase.

How does .NET MAUI work?



.NET MAUI timeline

https://github.com/dotnet/maui/wiki/roadmap

.NET 6 Preview 6

dotnet install

.NET 6 RC1

Android S & iOS 15 may be stable sometime in this timeframe

Bug Fixes

Ignite Conference

Jul 14, 2021 | Aug 10, 2021 | Sept 14, 2021 | Oct 12, 2021 | Oct 13, 2021 | Nov 9, 2021

We are here

.NET 6 Preview 7

VS 2022 Templates
C# Hot Reload
XAML Hot Reload
Hot Restart
iOS Remote Simulator
try-convert support

FEATURE COMPLETE

.NET 6 RC2

Bug Fixes

.NET 6 GA

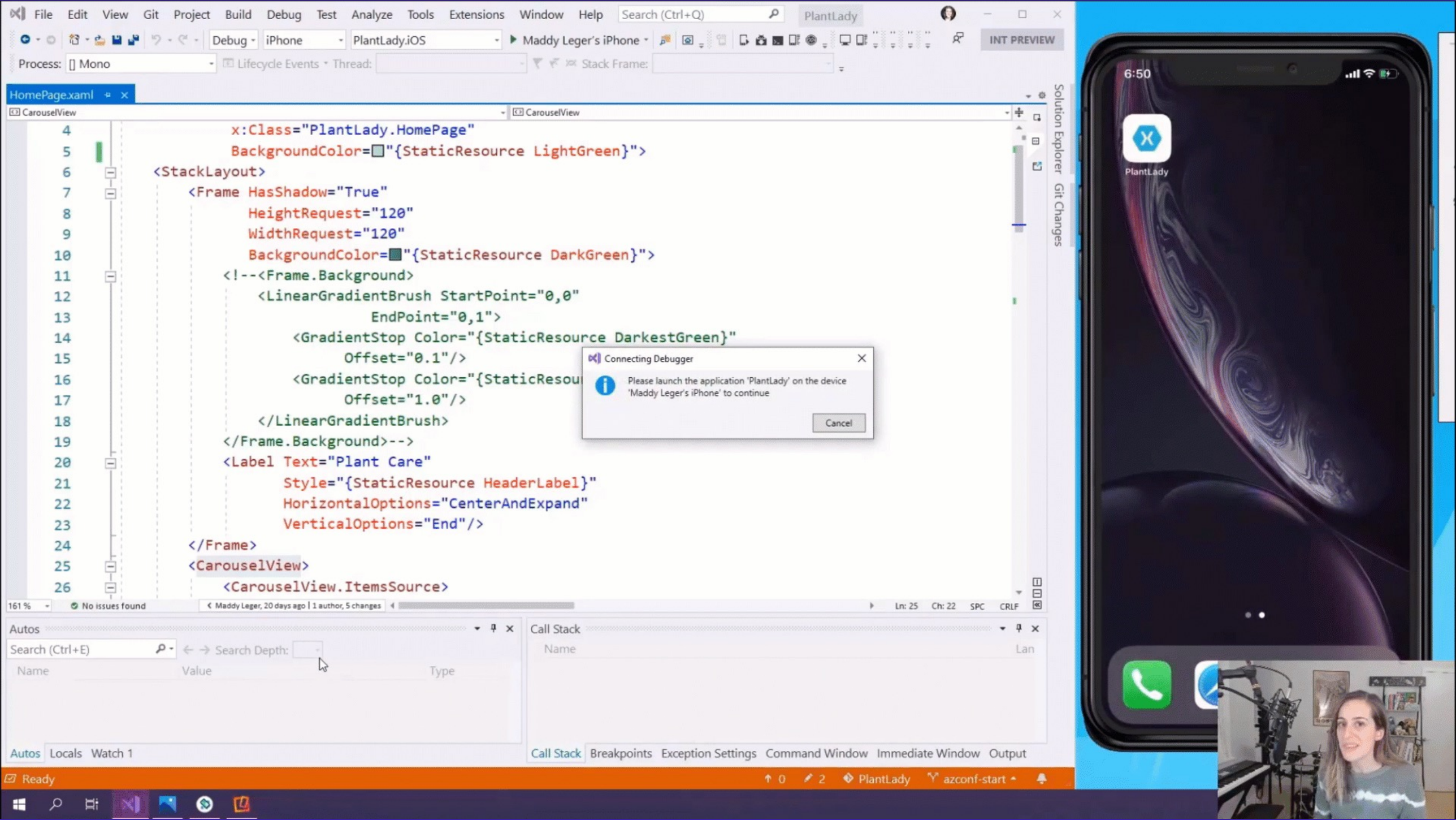


Let’s look at

- Install .NET 6 & MAUI
- Create a new .NET MAUI project
- New project features
- Run for multiple targets
- Coming Visual Studio 2022 Features

But wait, there's more!

- Develop iOS directly from your Windows machine (Hot Restart)
- Deploy Blazor apps on .NET MAUI
- Startup Profiling improves Android startup speed
- .NET MAUI includes everything in Xamarin and Xamarin.Forms, plus adds new features like Borders, Corners, Shadows for all platforms.
- 3rd party control vendors and libraries are adding .NET 6 support for MAUI


File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
Search (Ctrl+Q)
PlantLady
Debug iPhone PlantLady.iOS Maddy Leger's iPhone
INT PREVIEW
Process: [] Mono
Lifecycle Events Thread:
Stack Frame:
HomePage.xaml
CarouselView
CarouselView
4 x:Class="PlantLady.HomePage"
5 BackgroundColor="{StaticResource LightGreen}">
6 <StackLayout>
7 <Frame HasShadow="True"
8 HeightRequest="120"
9 WidthRequest="120"
10 BackgroundColor="{StaticResource DarkGreen}">
11 <!--<Frame.Background>
12 <LinearGradientBrush StartPoint="0,0"
13 EndPoint="0,1">
14 <GradientStop Color="{StaticResource DarkestGreen}"
15 Offset="0.1"/>
16 <GradientStop Color="{StaticResou
17 Offset="1.0"/>
18 </LinearGradientBrush>
19 </Frame.Background>-->
20 <Label Text="Plant Care"
21 Style="{StaticResource HeaderLabel}"
22 HorizontalOptions="CenterAndExpand"
23 VerticalOptions="End"/>
24 </Frame>
25 <CarouselView>
26 <CarouselView.ItemsSource>
Connecting Debugger
Please launch the application 'PlantLady' on the device 'Maddy Leger's iPhone' to continue
Cancel
161 %
No issues found
Maddy Leger, 20 days ago | 1 author, 5 changes
Ln: 25 Ch: 22 SPC CRLF
Solution Explorer Git Changes
Autos
Search (Ctrl+E)
Search Depth:
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Autos Locals Watch 1
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Call Stack Breakpoints Exception Settings Command Window Immediate Window Output
Ready
0 2 PlantLady azconf-start
6:50
PlantLady

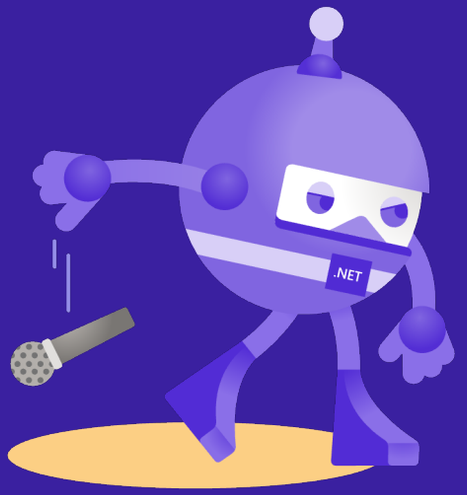
Summary

- .NET MAUI ships as part of **.NET 6** this November
- Use **.NET MAUI** to build native Android, iOS, macOS, and Windows applications
- **Single, multi-targeted project** helps you focus on your application more than different platforms
- **Visual Studio 2022** speeds up your development with Hot Reload and Hot Restart for iOS

Thanks for joining!

David Ortinau | Principal Program Manager | Microsoft

david.ortinau@microsoft.com | @davidortinau

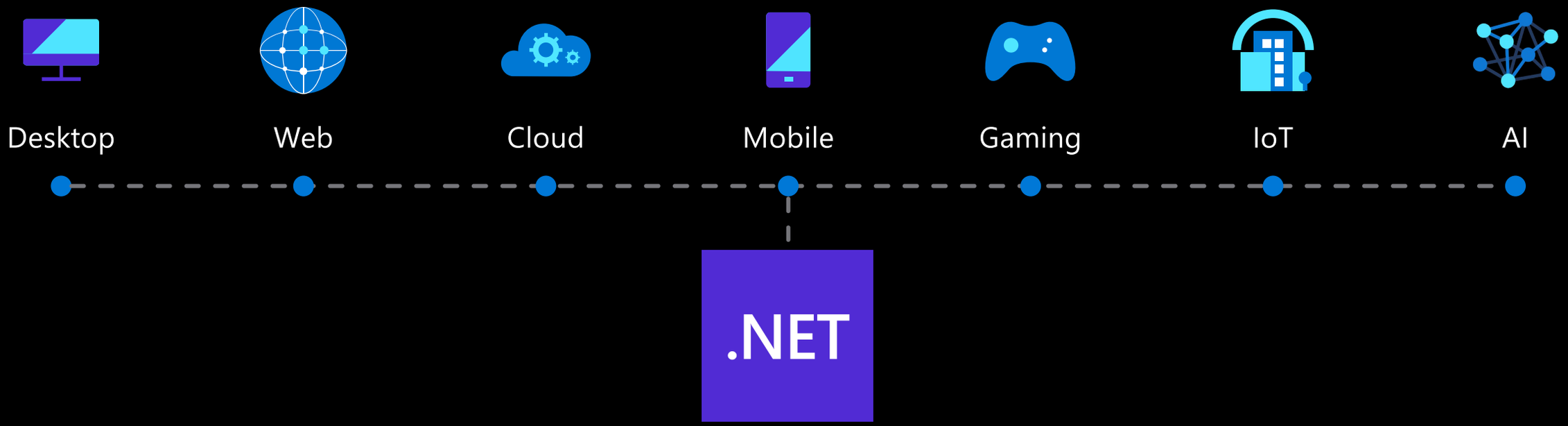


Microsoft

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.NET

Your platform for building anything



.NET ecosystem momentum

>5M

.NET developers

In Visual Studio family

#1

Most Loved Framework (2019 & 2020)

.NET Core

Top 30

Highest velocity OSS project

github.com/dotnet

Top 5

Language on GitHub

C#

>10X

Faster than Node.js

ASP.NET Core

40%

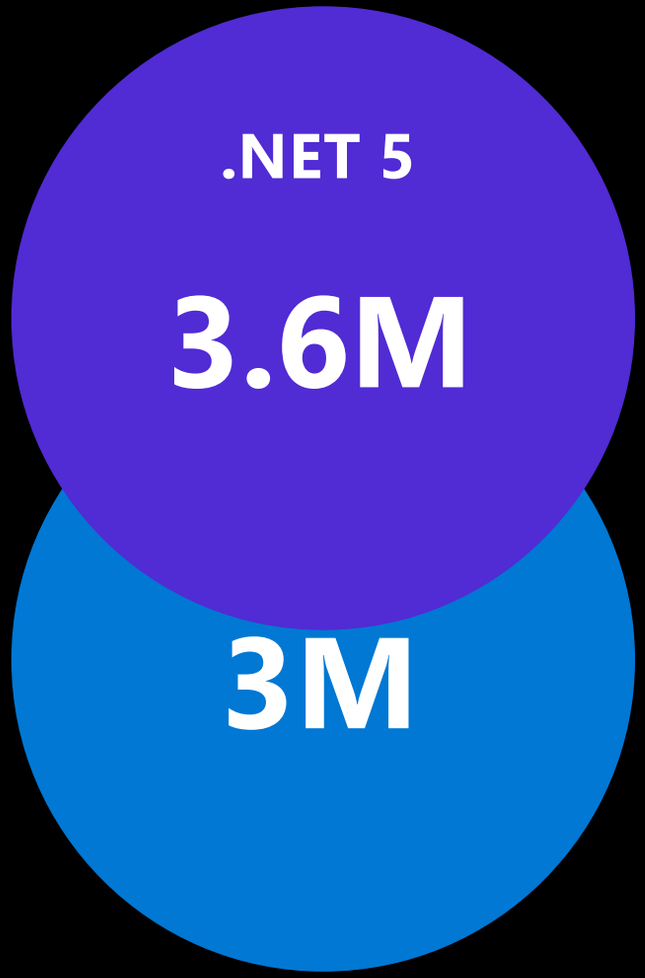
New to .NET are students

dot.net download survey

Sources: insights.stackoverflow.com/survey/2019#most-loved-dreaded-and-wanted; insights.stackoverflow.com/survey/2020#most-loved-dreaded-and-wanted; github.com/cncf/velocity/blob/master/reports/top_projects_by_activity.txt; octoverse.github.com/; www.techempower.com/benchmarks/#section=data-r20&hw=ph&test=plaintext

.NET adoption

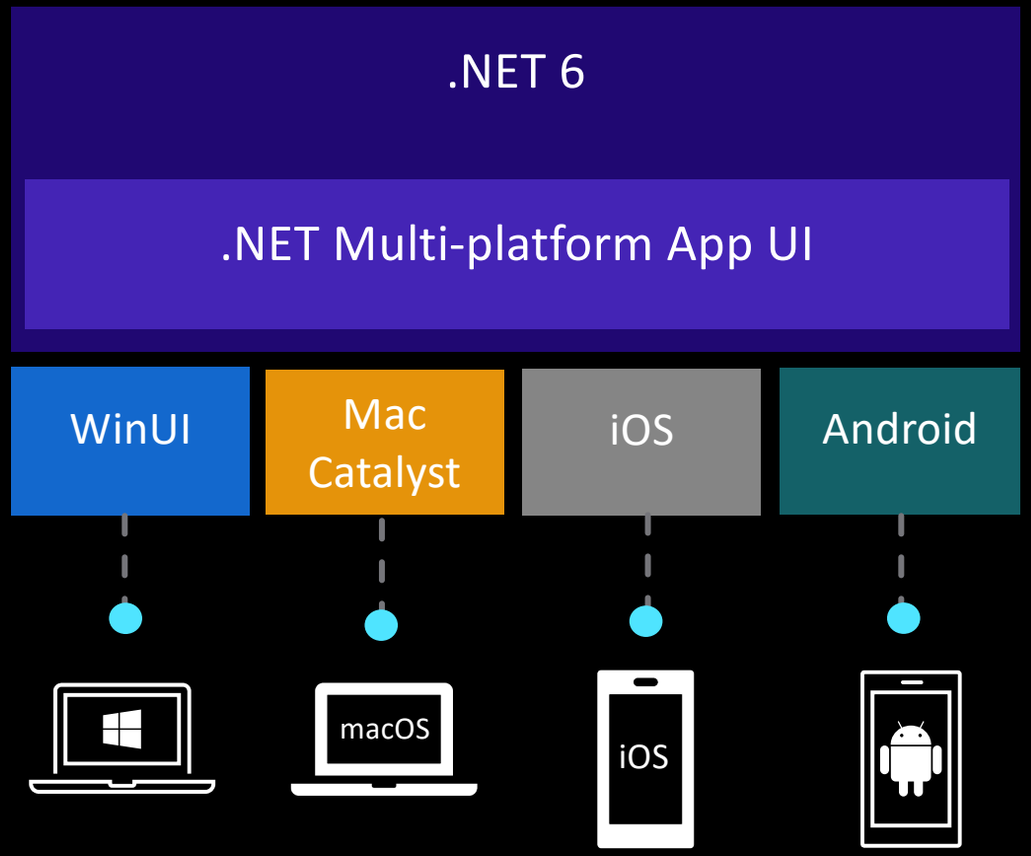
Downloads in first 6 months of release



Introducing .NET Multi-platform App UI



.NET Multi-platform App UI



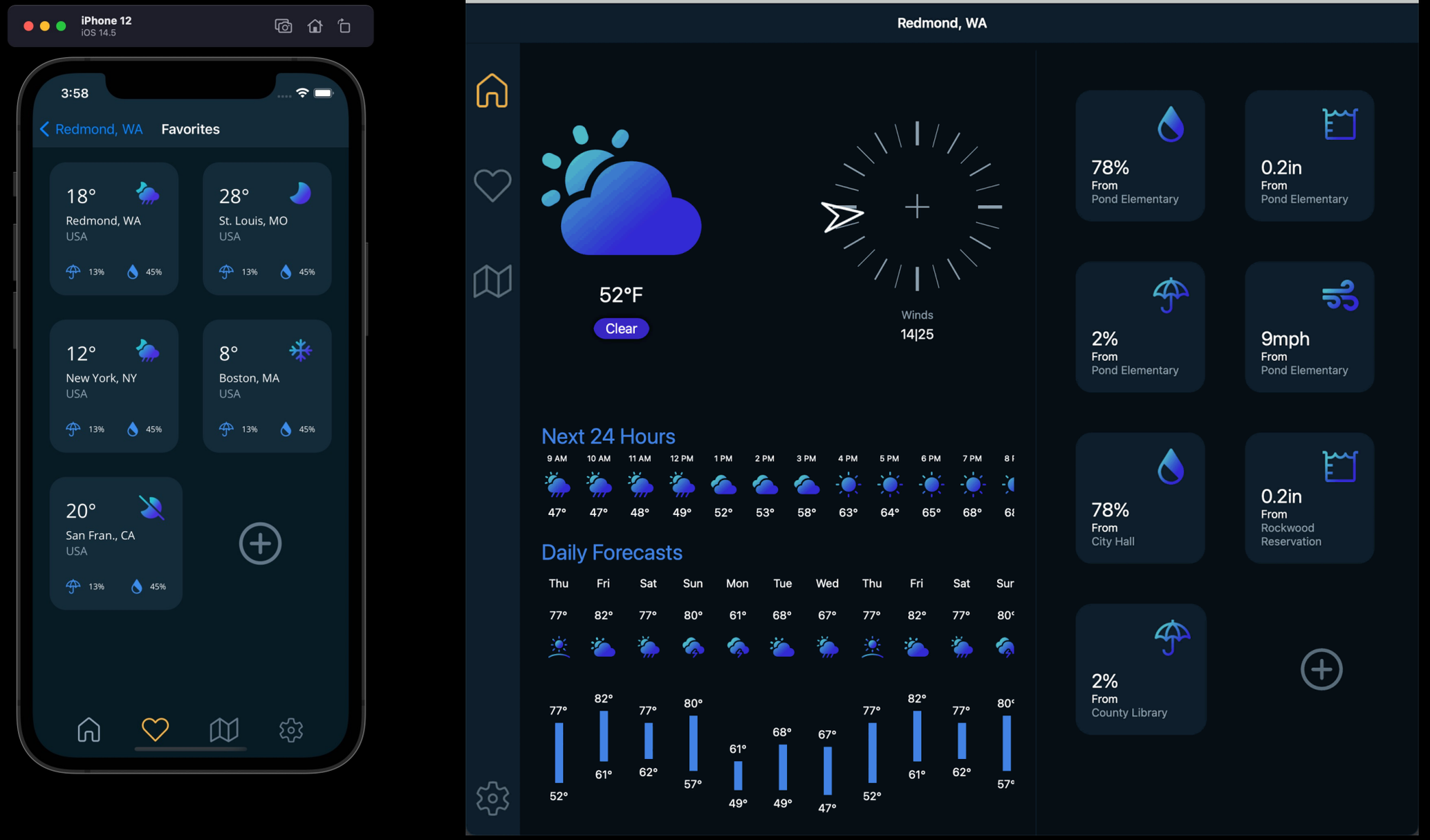
Cross-platform, native UI

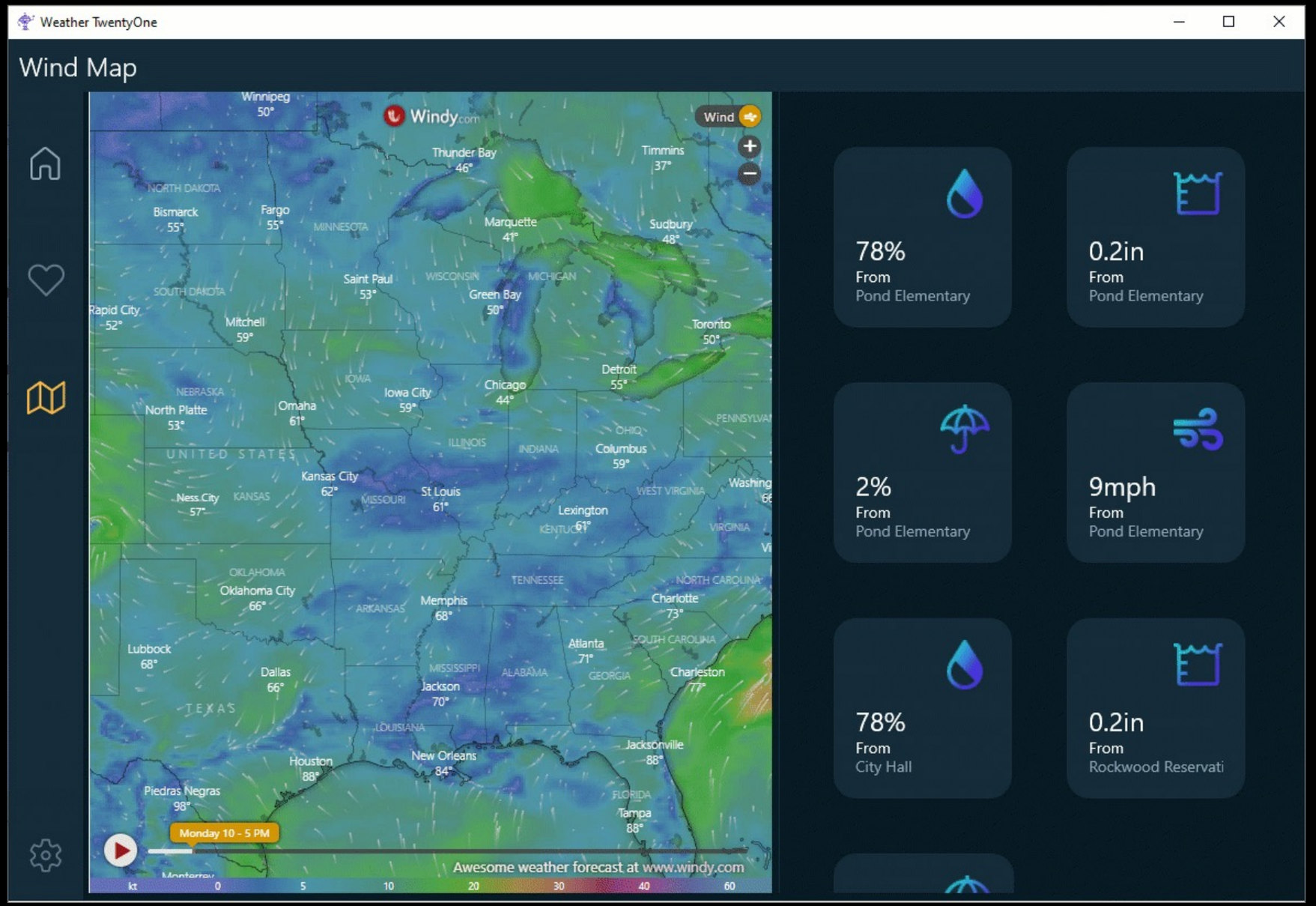
Single project system, single codebase

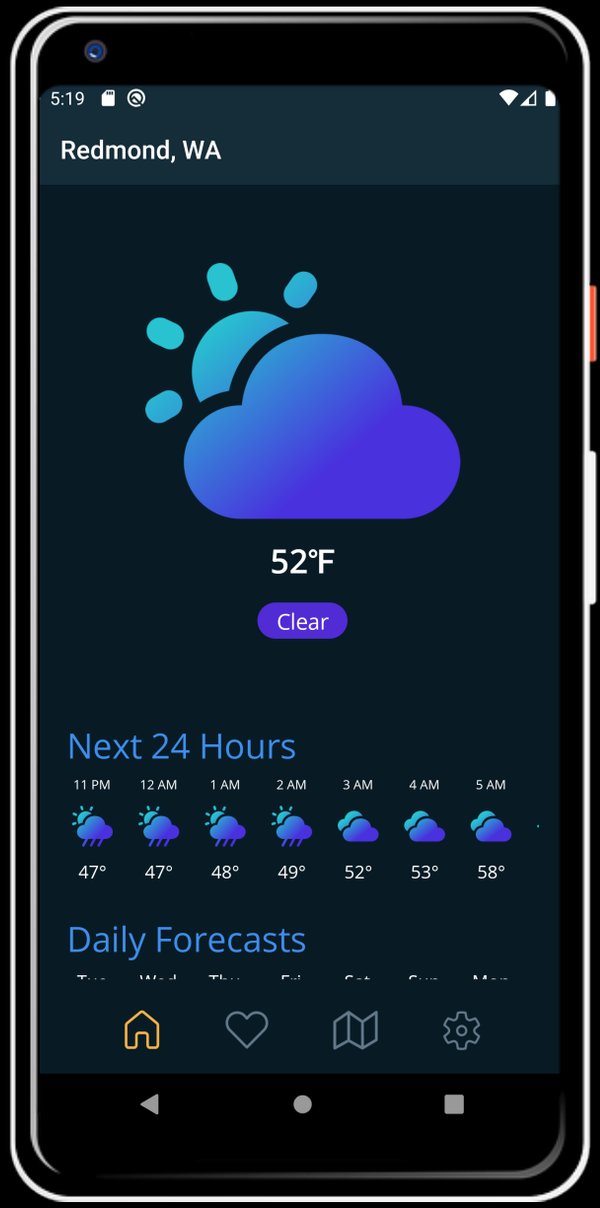
Deploy to multiple devices, mobile & desktop

General availability in .NET 6

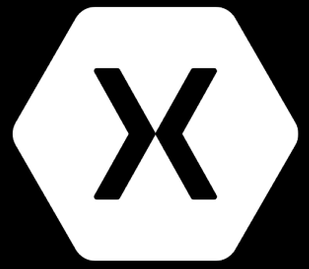
github.com/dotnet/maui


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Weather TwentyOne
Wind Map
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Monday 10 - 5 PM
Awesome weather forecast at www.windy.com
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Xamarin

Loved by developers, trusted by enterprise

xamarin.com/customers







































































.NET Conf 2020

November 10-12



Customer Showcase: The Acader

https://devblogs.microsoft.com/xamarin/customer-showcas...

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Customer Showcase: The Academy of Motion Picture Arts and Sciences



Alex

September 24th, 2020

The Academy of Motion Picture Arts and Sciences is known for its many areas of involvement in the movie industry, but it is most famous for the annual awards show, The Oscar's ®.

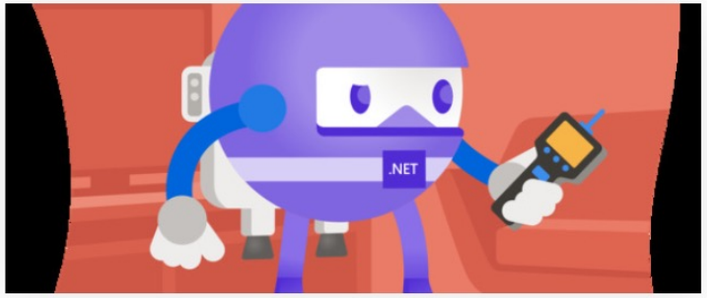
ACADEMY
OF MOTION PICTURE
ARTS AND SCIENCES

The development and infrastructure teams needed to update how The Academy's members interacted with their events, managed their profiles, and viewed the nominated movies or short films. They knew they could count on Azure, Visual Studio, .NET, and Xamarin to help deliver their vision.

Throughout the year, The Academy moved their asp.net applications and backend services, built with .NET framework 4.x, from their own data centers into Microsoft Azure. All of this requiring little-to-no code changes. This has allowed their development team to build new iOS, Android, and tvOS applications with Xamarin.Forms. In turn, delivering a great experience for the Academy members.

> "When I was first asked to investigate an approach for building a tvOS app several years ago, I started down the Swift route. When revisiting the topic sometime later, I saw how much Xamarin had matured. Today, native bindings for Xamarin are updated quickly, typically within a few days. So that everything pretty much works like a native SDK library."
> **– Nick Amperano, Director Of Engineering**

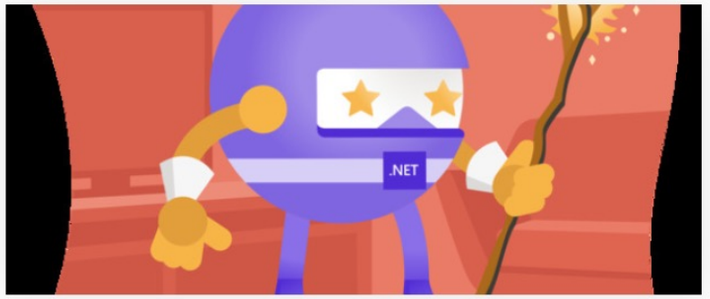
Feedback



Episode 91: LivePerson

April 23rd, 2021 | 37 mins 2 secs

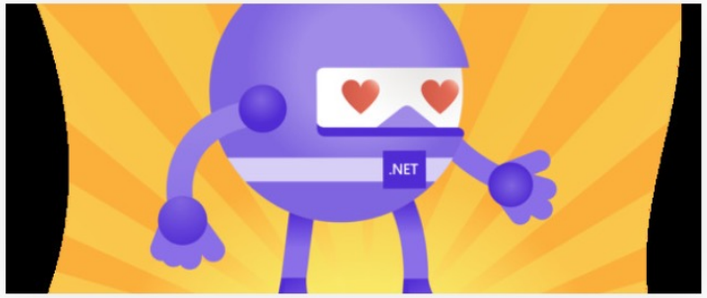
android, ios, uwp, xamarin, xamarin.forms



Episode 78: The Oscars

September 22nd, 2020 | 42 mins 44 secs

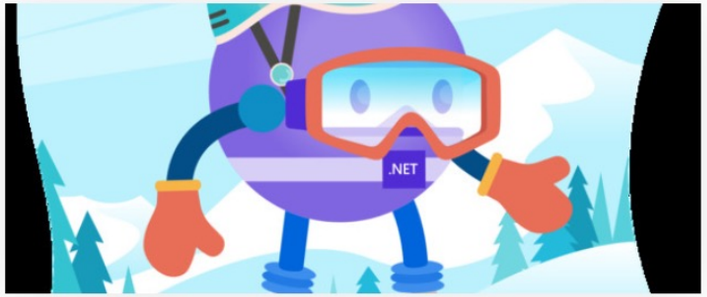
android, ios, uwp, xamarin, xamarin.forms



Episode 82: iRel8

November 20th, 2020 | 51 mins 24 secs

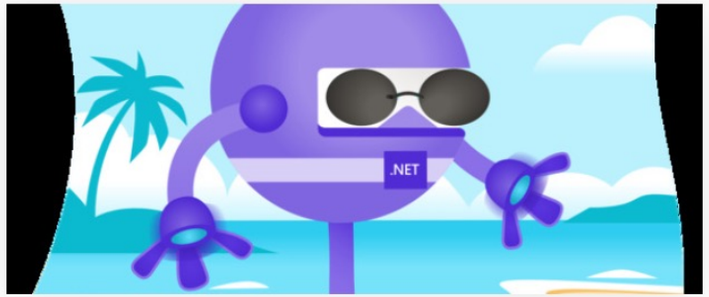
android, ios, uwp, xamarin, xamarin.forms



Episode 89: Polaris

March 26th, 2021 | 44 mins 24 secs

android, ios, uwp, xamarin, xamarin.forms



Episode 84: American Airlines

December 18th, 2020 | 32 mins

android, ios, uwp, xamarin, xamarin.forms



Episode 86: Medusa

January 22nd, 2021 | 43 mins 2 secs

android, ios, uwp, xamarin, xamarin.forms



XamarinPodcast.com

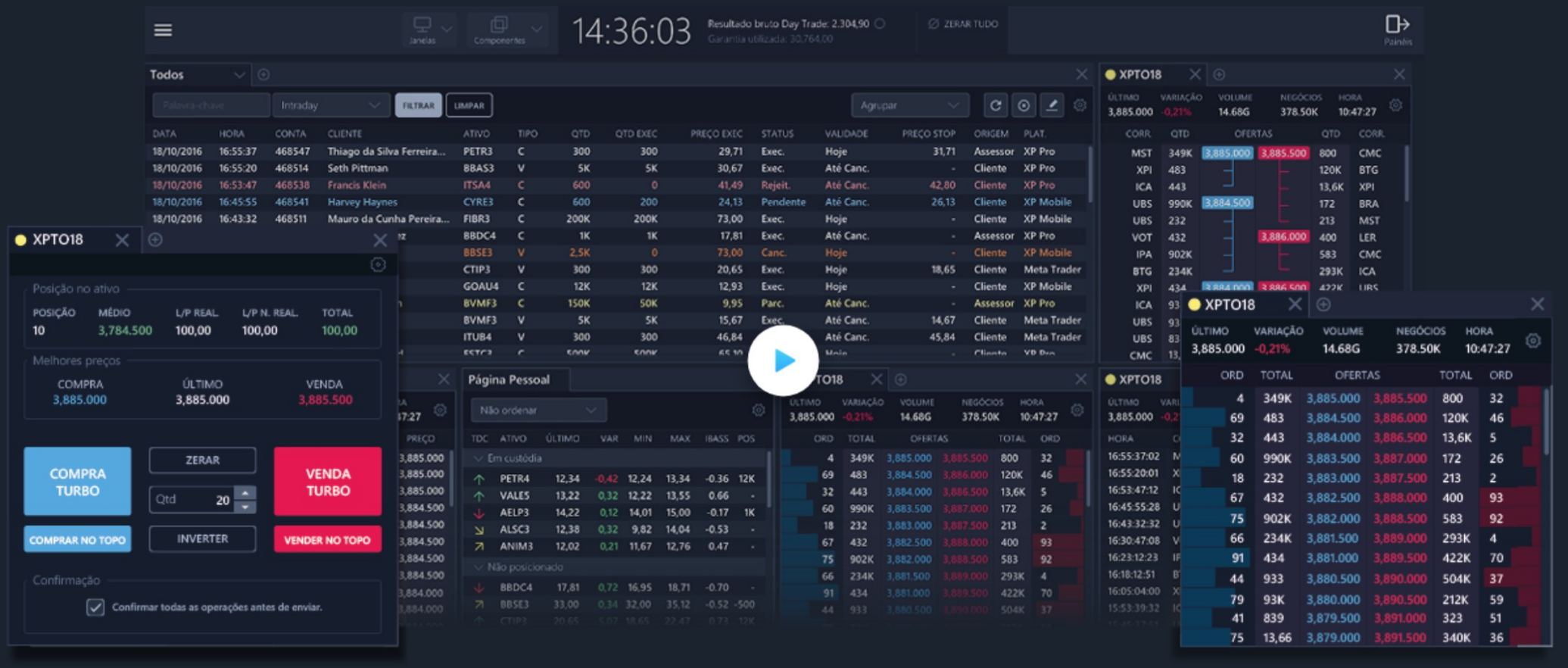
BULL Tecfinance

https://www.bulltf.com.br

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Conheça o BULL e nos ajude a transformar a forma de operar na Bolsa.

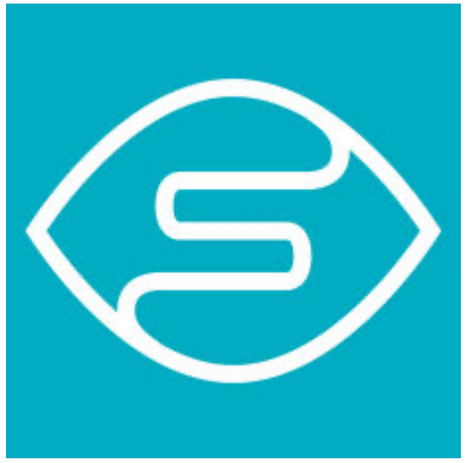


First Party Apps



Microsoft News

Using the Microsoft News app, you'll find best-in-class content from market leaders around the world. Automatically sync your favorite topics across Microsoft News on the web and your mobile apps for quick access to the news you want to read.



Seeing AI

An app that narrates the world around you. Designed for the low vision community, this research project harnesses the power of AI to describe people, text and objects



Azure App

With the Azure mobile app, you don't need to be in front of your computer to keep an eye on your Azure resources such as VMs and web apps. Stay connected no matter where you are from your iOS or Android mobile device.



O365 Admin

Are you a Microsoft 365 admin? Then you need the Microsoft 365 Admin app. Manager users, reset passwords, assign licenses, control notifications for service alerts, and connect with Microsoft Support.

Xamarin.Forms + Azure

Governments, healthcare, and others rapidly create safe and secure apps to address the COVID-19 crisis.

An open-source contact tracing app for Japanese Government.

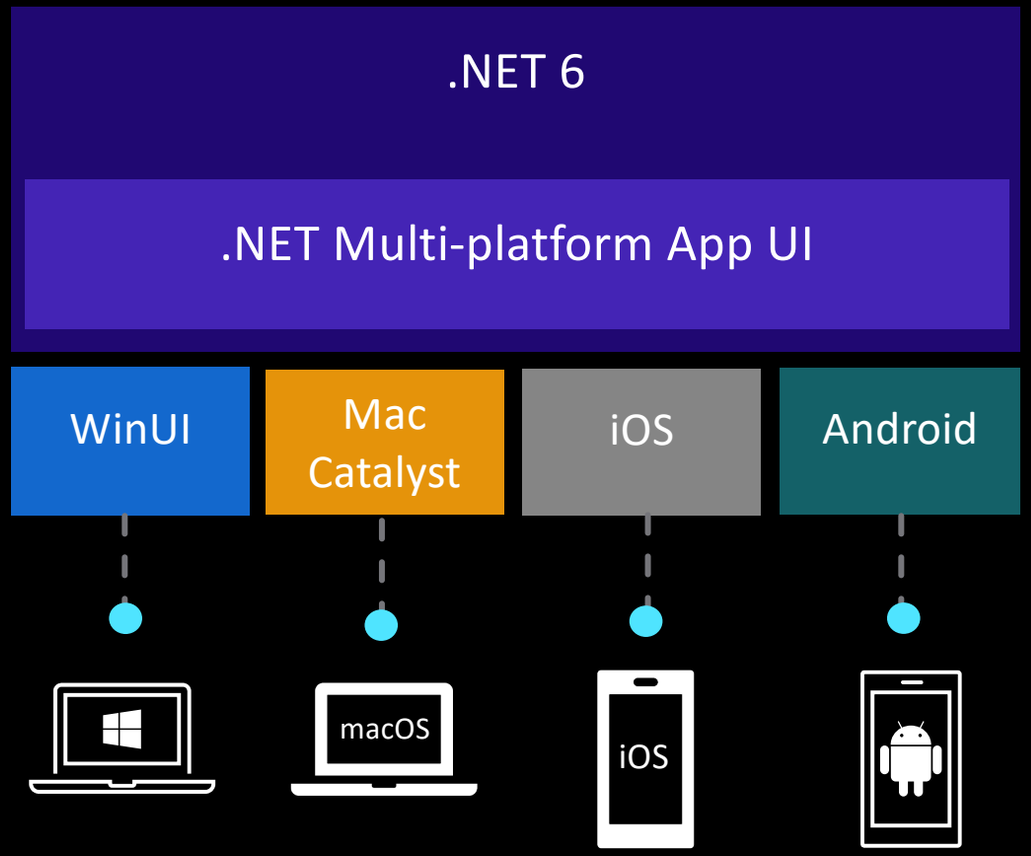
https://github.com/Covid-19Radar/Covid19Radar



The Roadmap to .NET 6



.NET Multi-platform App UI



Cross-platform, native UI

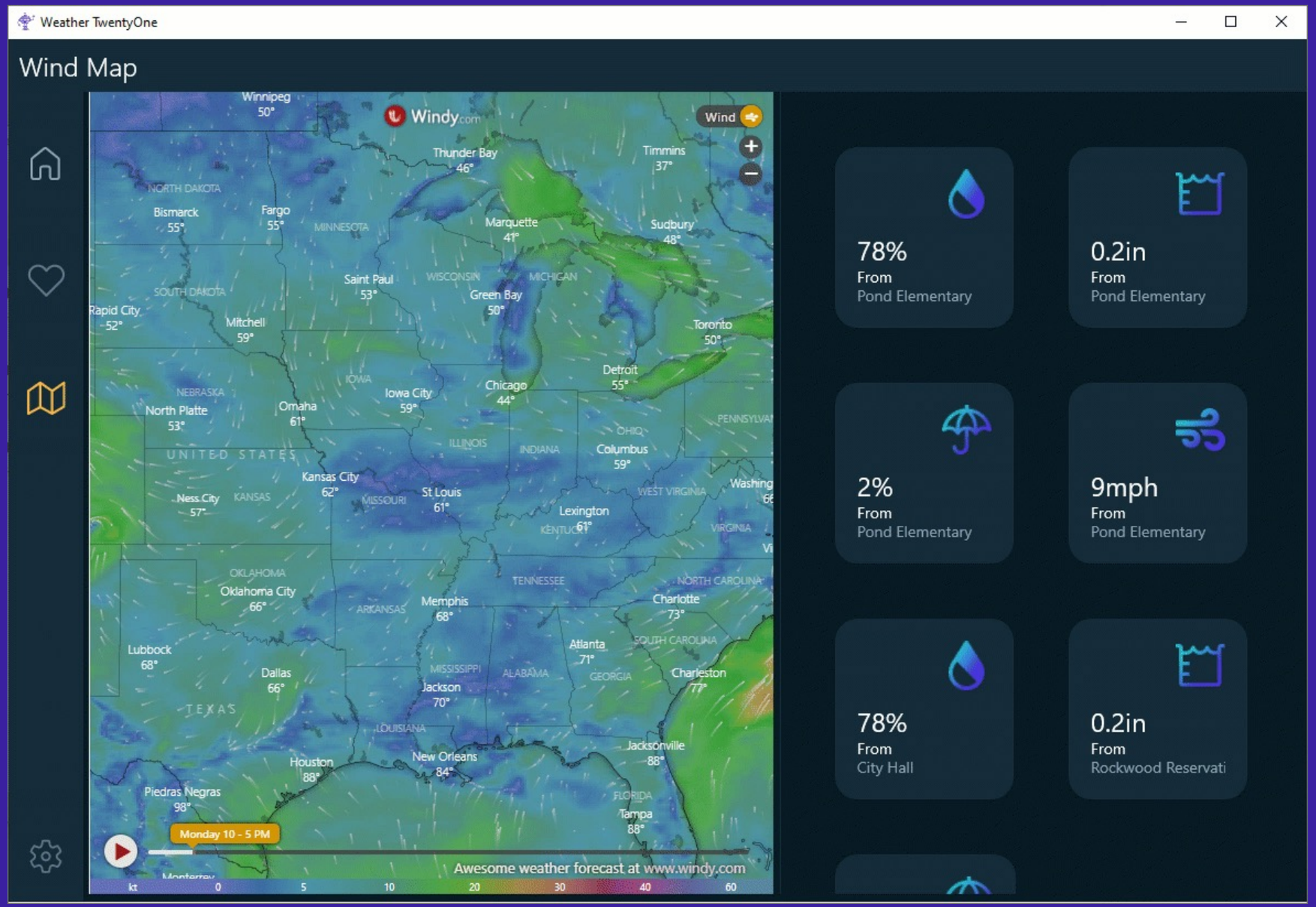
Single project system, single codebase

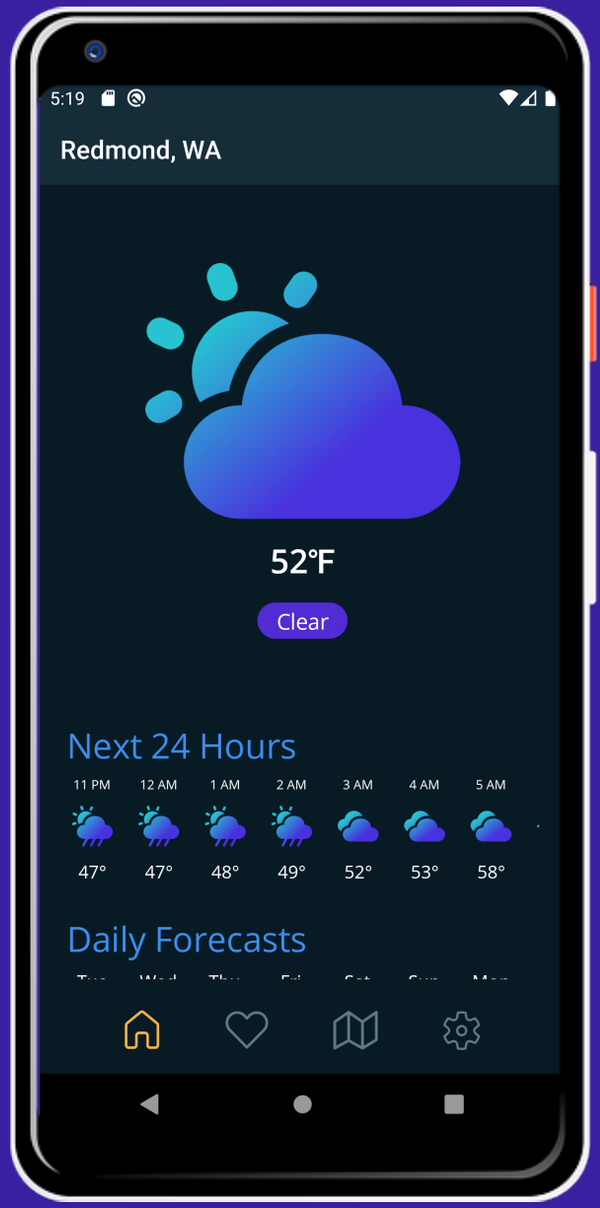
Deploy to multiple devices, mobile & desktop

General availability in .NET 6

github.com/dotnet/maui


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.NET MAUI

Microsoft.Maui *evolution of Xamarin.Forms & Xamarin.Essentials*

File | New

Multi-platform App UI (.NET)

CLI

dotnet new maui



.NET for iOS and Android

Xamarin.iOS --> .NET for iOS
Xamarin.Android --> .NET for Android

Namespaces & APIs unchanged from existing UIKit & Android APIs

File | New (SDK style)

CLI

dotnet new ios

dotnet new android

What’s Improved in .NET MAUI

Control architecture

Native + Drawn UI

Layout Manager

App startup

Multi-window support

Borders, Corners, & Shadows

.NET Hot Reload

Single Project

- Multi-targeting
- Image, font, raw media management
- Streamlined project settings

Design Themes

UI Styling in .NET MAUI

Consistent design by default

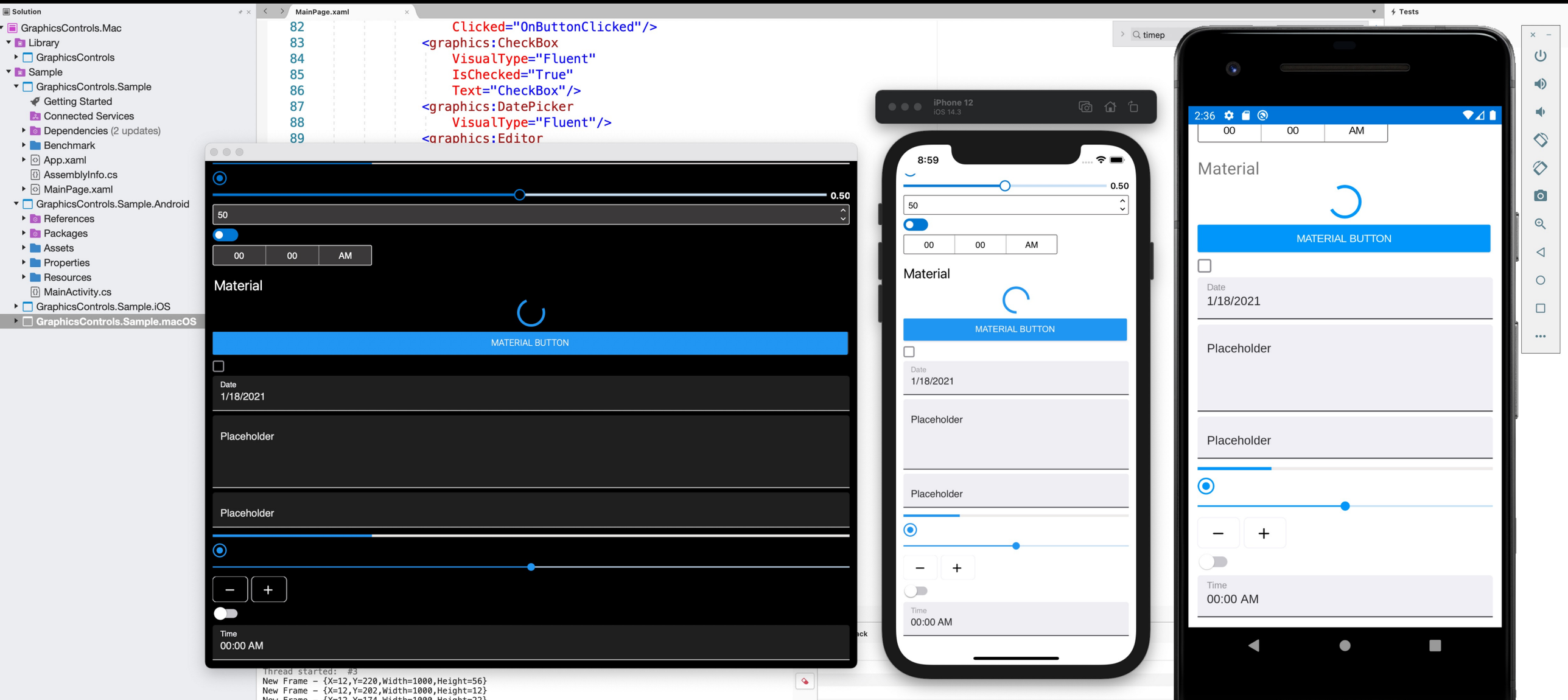
Native + Drawn controls

Supporting themes:

- Fluent UI
- Material Design
- Platform Default

Drawn Controls

https://github.com/dotnet/Microsoft.Maui.Graphics.Controls

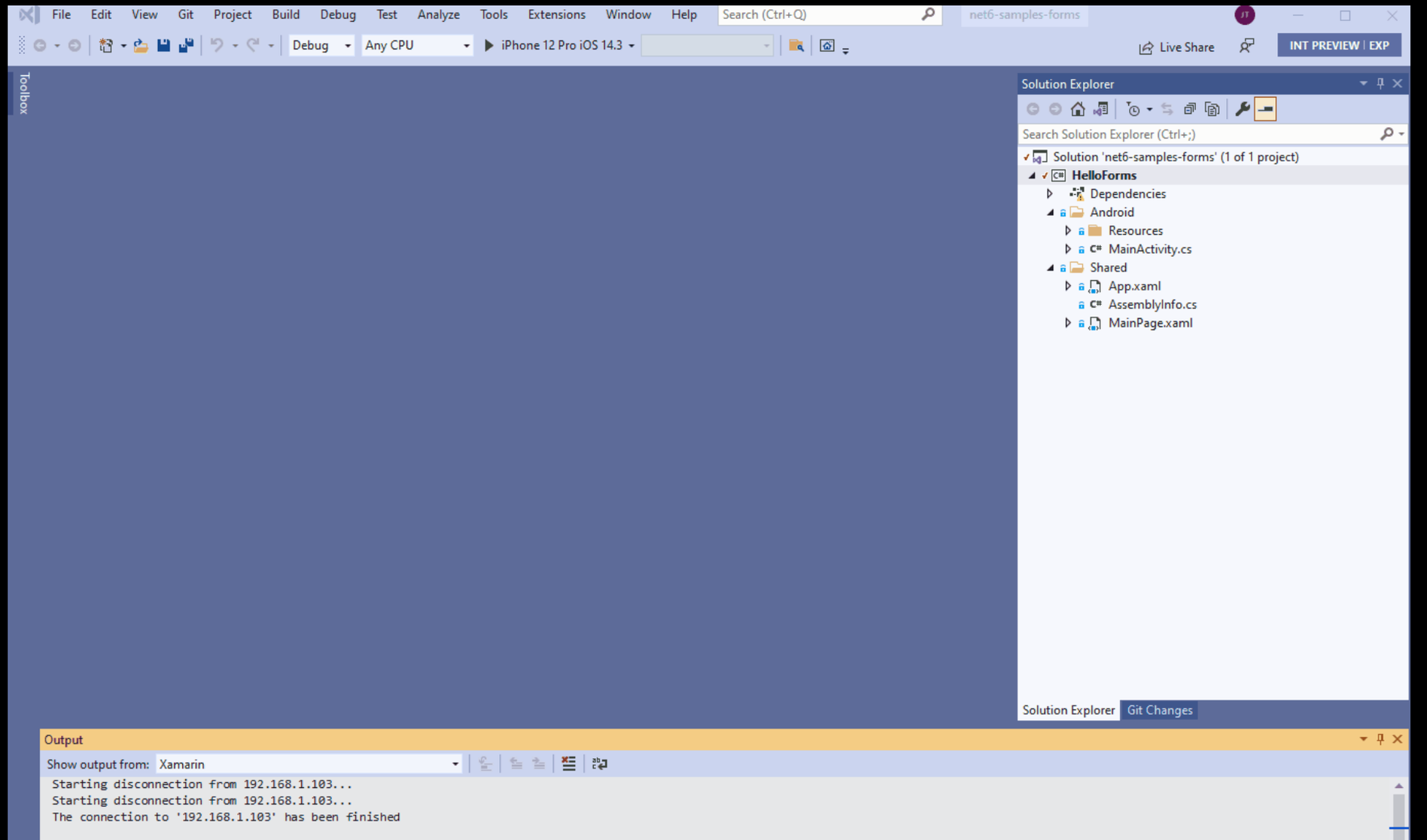


Developer Experience

Simplified Project Solution
Hot Reload for C#, Razor, and XAML
Improved Debugging



Simplified Cross-Platform Development



Single place for:

- Images/Assets
- App icons
- Fonts
- Launch screens
- Manifest
- Startup/App.cs

...and more!

Deeper: MAUI

Handlers
Lifecycle

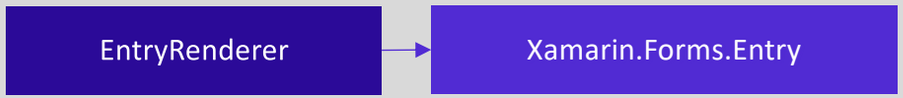


Evolving the Architecture

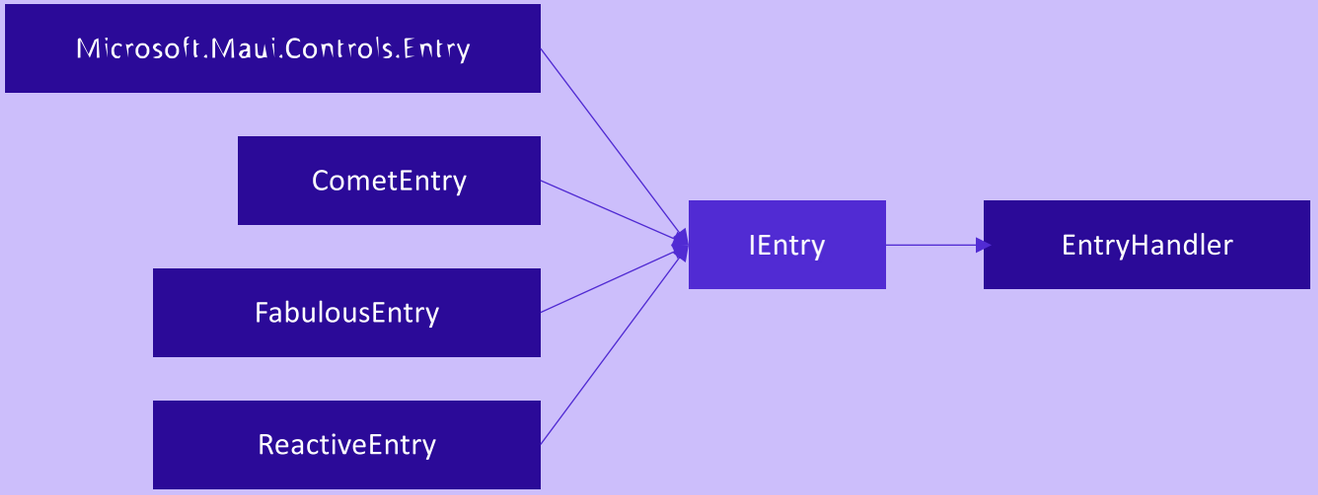
Renderers are tightly coupled to Xamarin.Forms and Xamarin.Forms components

Handlers are unwrapped, decoupled, portable, and reusable

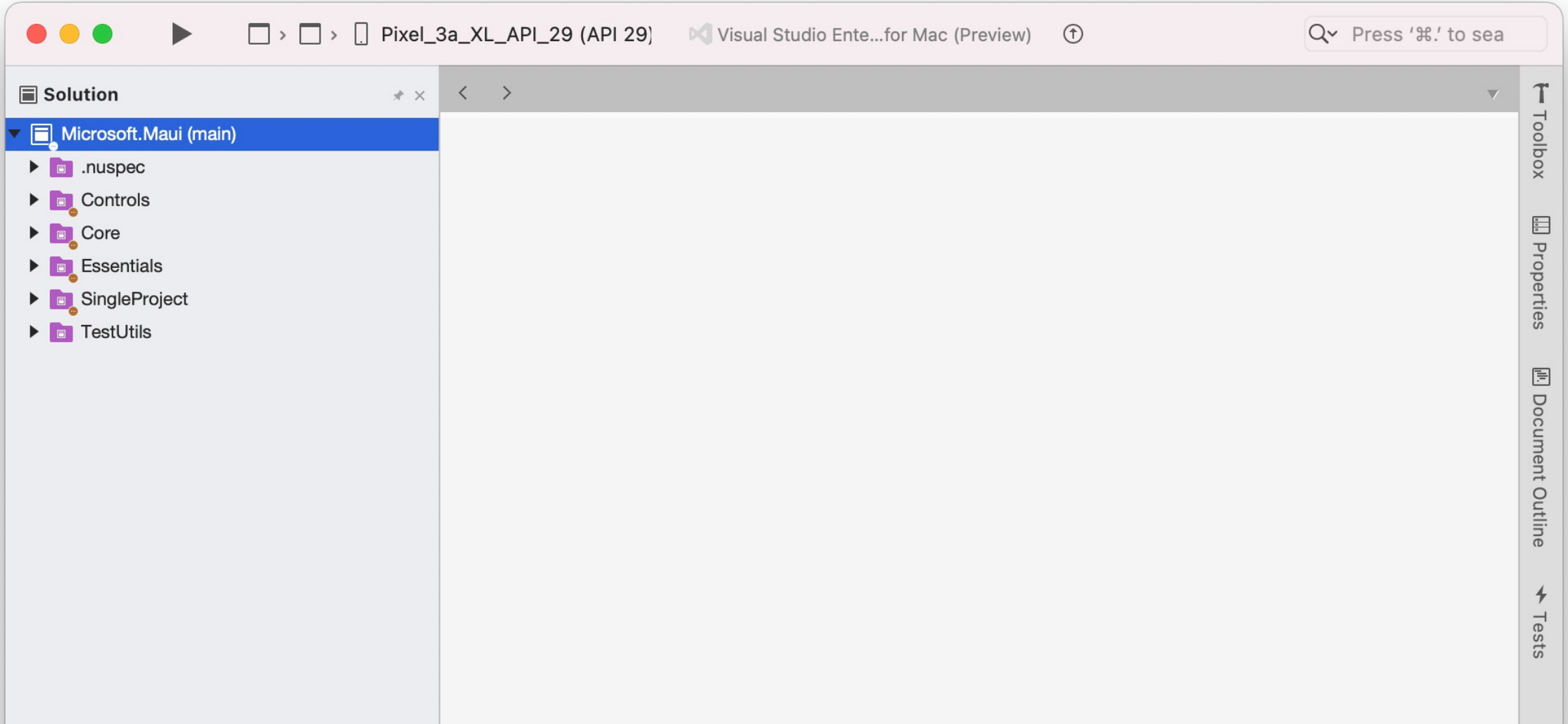
OLD



NEW



Entry : IEntry



Extending Controls

“So…if I’m not creating Custom Renderers, Effects, and PlatformSpecifics anymore…

how do I _____?”

Goodbye, Android Entry Underline

```
#if ANDROID
Handlers.EntryHandler
    .EntryMapper[nameof(IEntry.BackgroundColor)] = (h,v) =>
    {
        (h.NativeView as global::Android.Views.Entry).UnderlineVisible = false
    };
#endif
```

#ifdef if platform specific

Access the handler from anywhere

Dictionary of all props

Debug Rainbows

```
Handlers.ViewHandler
    .ViewMapper[nameof(IView.BackgroundColor)] = (h, v) =>
    {
       (h.NativeView as global::Android.Views.View).BackgroundColor
               = RandomColor();
    };
```

Make your a11y testing life easier!

```
#if ENABLE_TEST_CLOUD
Handlers.ViewHandler
    .ViewMapper[nameof(IView.AutomationId)] = (h, v) =>
    {
        (h.NativeView as global::Android.Views.View).ContentDescription
            = v.AutomationId;
    };
#endif
```

Lifecycle

Cross Platform Lifecycle

The Application, Window, Page and View classes contains virtual methods that can be overridden to respond to lifecycle changes.

```
protected override void OnCreated()
{
    Debug.WriteLine ("OnCreated");
}
protected override void OnResumed()
{
    Debug.WriteLine ("OnResumed");
}
protected override void OnPaused()
{
    Debug.WriteLine ("OnPaused");
}
```

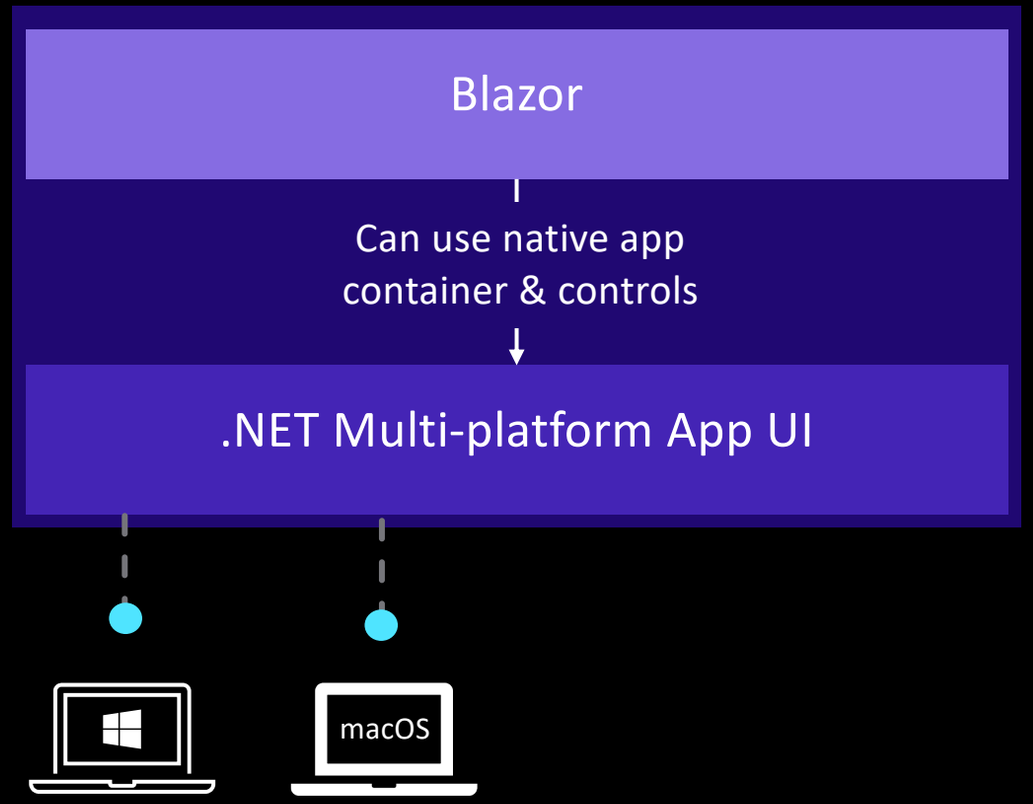
Native Platform Lifecycle

.NET MAUI also will allow to hook into native life cycle events without the complexities of wiring up a renderer so if require finer platform level hooks they can hook in in an easy way.

```
appBuilder.ConfigureAndroidLifecyle(lifecycle => {
              lifecycle
                            .OnCreate((ctx, state) => { AppCenter.Init(ctx, state);  })
                            .OnStart(() => { AppCenter.StartTracking(); });
});
```

https://github.com/dotnet/maui/issues/30

Blazor hybrid desktop apps



Reuse UI components across native and web

Built on top of .NET Multi-platform App UI

Native app container & embedded controls

General availability in .NET 6

Release Schedule

aka.ms/xf-roadmap

Jan 2021	Xamarin.Forms 5.0
Jan-Nov 2021	Xamarin will support new OS versions - Services releases as needed
Nov 2021	.NET 6 GA
Nov 2022	Xamarin supported completes

Microsoft

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